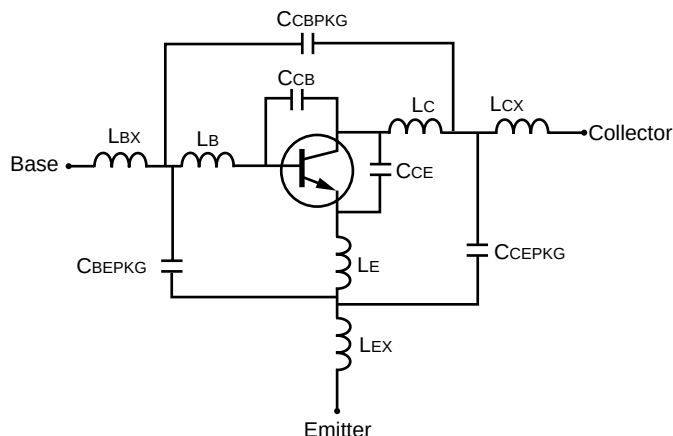


SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS ⁽¹⁾

Parameters	Q1	Parameters	Q1
IS	2.2e-18	MJC	0.33
BF	120	XCJC	1
NF	1	CJS	0
VAF	39.7	VJS	0.75
IKF	0.21	MJS	0
ISE	4.64e-14	FC	0.5
NE	2.09	TF	2e-12
BR	10	XTF	20
NR	1.004	VTF	10
VAR	1.9	ITF	0.1
IKR	0.1	PTF	200
ISC	1.1e-11	TR	1e-11
NC	41	EG	1.11
RE	3.5	XTB	0
RB	14	XTI	3
RBM	14	KF	0
IRB	0.004	AF	1
RC	8		
CJE	0.3e-12		
VJE	0.5		
MJE	0.33		
CJC	0.001e-12		
VJC	0.75		

(1) Gummel-Poon Model

ADDITIONAL PARAMETERS

Parameters	NE661M04
CCB	0.08e-12
CCE	0.1e-12
LB	0.93e-9
LC	0.6e-9
LE	0.2e-9
CCBPKG	0.001e-12
CCEPKG	0.25e-12
CBEPK	0.2e-12
LBX	0.2e-9
LCX	0.2e-9
LEX	0.05e-9

MODEL RANGE

Frequency: 0.1 to 18 GHz
 Bias: $V_{CE} = 0.5 \text{ V to } 2.5 \text{ V}$, $I_C = 1 \text{ mA to } 7 \text{ mA}$
 Date: 02/2002

Life Support Applications

These NEC products are not intended for use in life support devices, appliances, or systems where the malfunction of these products can reasonably be expected to result in personal injury. The customers of CEL using or selling these products for use in such applications do so at their own risk and agree to fully indemnify CEL for all damages resulting from such improper use or sale.